Introduction to Algorithms



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Learning Outcomes

- On completion of this session, you should be able to
  - Define an algorithm
  - Identify the properties of a good algorithm
  - Define a program
  - Identify the properties of a good program
  - Specify algorithms in pseudocode
  - Explain the efficiency of an algorithm
  - Distinguish different approaches used to determine the efficiency of algorithms

Algorithm

- What's an algorithm?
  - A recipe
  - A finite, well-defined set of steps for solving a problem
  - Can be more than one algorithms to solve a problem
- A good algorithm should
  - be correct
  - be composed of a finite number of well-defined steps
  - produce output
  - be able to terminate in finite time
  - be efficient

Program

- What's a program?
  - An implementation of an algorithm using a programming language
- Good program should be
  - able to execute correctly
  - able to execute efficiently (within the resource constraints)
  - easy to read and understand
  - easy to maintain

Pseudocode

- A kind of structured English for specifying algorithms
- Means fake code
- Can not be compiled nor executed
- No strict syntax rules
- Easy to read and understand
- Programming language independent
- Designed for humans, not machines

Pseudocode

- Example

```
ALGORITHM search (REF list[] INTEGER, size INTEGER,
   target INTEGER, REF index INTEGER)
   finds target in unordered list
   PRE    list is an array with at least one element
          size is number of elements in list
          target contains integer to find
   POST   if found, index gives matching index
   RETURN true if found, otherwise false
1  index ← 0
2  LOOP (index < size)
3     IF (target = list[index])
4        RETURN true
5     END IF
6     index ← index + 1
7  END LOOP
8  RETURN false
END search
```

Efficiency of an Algorithm

- Amount of memory and time needed to run a program written with a particular algorithm
- Used to pick the most appropriate one when there are multiple algorithms exist for the same problem
- Determining efficiency
  - Performance measurement
  - Performance analysis

Questions